IO391
Digital I/O Module
User Manual

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1 Introduction

The IO391 is a standard full PCI Express Mini Card, providing a user programmable Xilinx Artix-7 7A50T FPGA.

The IO391 provides 26 ESD-protected 5 V-tolerant TTL lines. All I/O lines are individually programmable as input or output. Each TTL I/O line has a pull-resistor to a common programmable pull voltage that can be set so +3.3 V, +5 V and GND.

The I/O signals are accessible through a 30 pin Pico-Clasp latching connector.

The User FPGA is configured by a SPI flash. An in-circuit debugging option is available via a JTAG header for read back and real-time debugging of the FPGA design (using the Vivado ILA). With the TA308 Programming Kit direct JTAG access to the FPGA is possible, using the Xilinx Platform Cable USB.

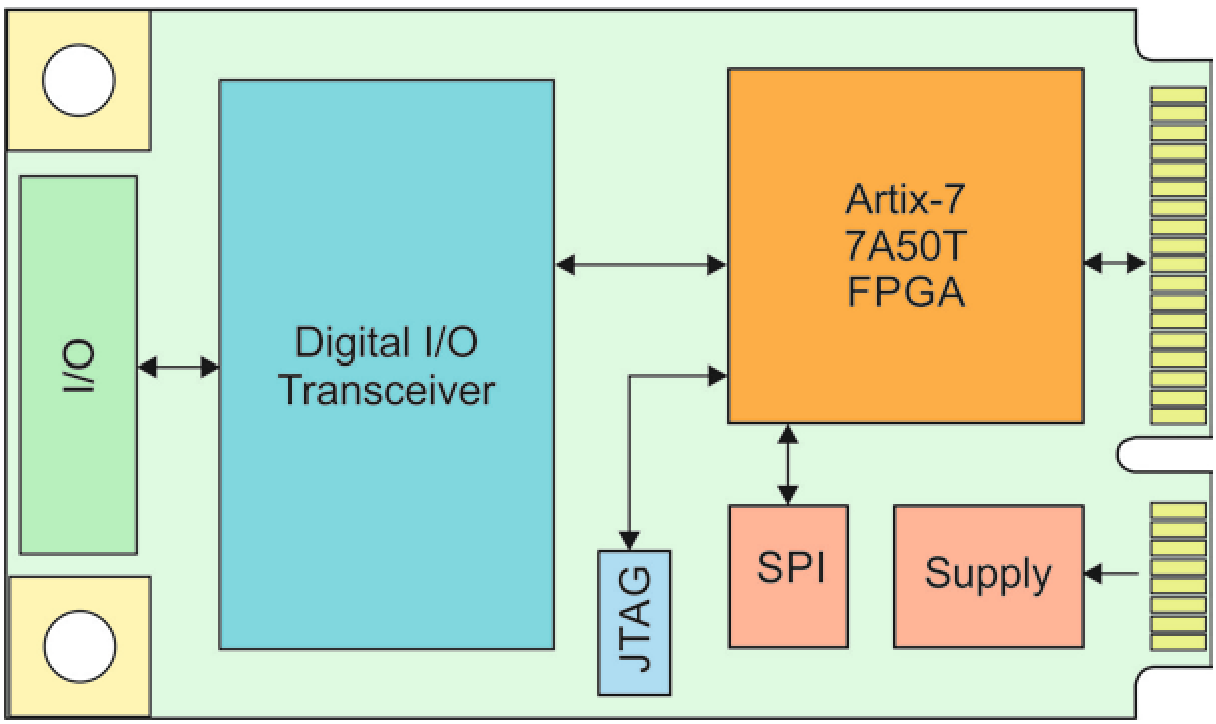


Figure 1-1: Block Diagram

2 Handling and Operating Instructions

2.1 ESD Protection

The PCI Express Mini Card module is sensitive to static electricity. Packing, unpacking and all other module handling has to be done in an ESD/EOS protected Area.

2.2 Height Restrictions

The I/O connector and the heatsink will exceed the available PCI Express Mini Card components height. Check carefully if your application provides enough spacing for an IO391.

2.3 Thermal Considerations

Due to its small size and high density, the IO391 can generate a lot of heat.

3 Technical Specification

Interface		
Mechanical Interface	PCI Express Mini Card conforming to PCI Express Mini Card Electromechanical Specification, Revision 2.0 Card Type: Full-Mini Card (50.95 x 30 mm)	
Electrical Interface	PCI Express x1 Link conforming to PCI Express Base Specification, Revision 2.0 The IO391 does not support the USB interface	
Main On-Board Devices		
User configurable FPGA	IO391: XC7A50T-2 (Xilinx)	
SPI-Flash	Macronix 128 Mbit SPI-Flash	
I/O Interface		
Digital I/O Channels	26 ESD-protected 5 V-tolerant TTL lines	
Digital I/O Transceiver	74LVC2G241 (or compatible)	
I/O Connector	30 pol. Pico-Clasp latching connector	
Physical Data		
Power Requirements	+3.3 Vaux: 650 mA typical	
Temperature Range	Operating	refer to “Thermal Management” chapter
	Storage	-40°C to +85°C
MTBF	987.000 h MTBF values shown are based on calculation according to MIL-HDBK-217F and MIL-HDBK-217F Notice 2; Environment: G_B 20°C. The MTBF calculation is based on component FIT rates provided by the component suppliers. If FIT rates are not available, MIL-HDBK-217F and MIL-HDBK-217F Notice 2 formulas are used for FIT rate calculation.	
Humidity	5 – 95 % non-condensing	
Weight	13 g	

Table 3-1: Technical Specification

4 Functional Description

This chapter gives a brief overview of the various module functions.

4.1 User FPGA Overview

The FPGA on the IO391 is an Artix-7 in a CSG325 package.

Artix-7	Slices	Flip-Flops	DSP48A1 Slices	Block RAM (Kb)	GTP Transceivers
7A15T	8150	65200	120	2700	4

Table 4-1: IO391 FPGA Feature Overview

4.1.1 Configuration via JTAG

Advanced users can indirectly program the SPI-Flash via the JTAG Chain. Additionally, the JTAG chain may be used with Xilinx Chipscope for debugging. See Xilinx documentation.

4.2 Clocking

4.2.1 FPGA Clock Sources

The following table lists the available clock sources on the IO391:

FPGA Clock-Pin Name	FPGA Pin Number	Source	Description
MGTREFCLK0_101	D5 / D6	PCI Express Mini Card Slot	100 MHz PCIe Reference clock
IO_L12P_T1_MRCC_14 P14	P14	External oscillator	100MHz External user clock
IO_L3N_T0_DQS_EMCCLK_14	K18	External oscillator	100 MHz External master configuration clock. The oscillator is not connected to a clock capable pin, and should not be used in designs for anything other than configuration.

Table 4-2: Available FPGA clocks

4.3 Digital I/O Interface

Each of the 14 digital I/O channels provides an input and output data signal and an output enable signal to the digital buffers.

Signal Name	Pin Number	Direction	IO Standard	IO Bank	Drive [mA]	Slew Rate
DI<0>	D14	IN	LVCMOS33	14	4	SLOW
DI<1>	D13	IN	LVCMOS33	14	4	SLOW
DI<2>	B14	IN	LVCMOS33	14	4	SLOW
DI<3>	G16	IN	LVCMOS33	14	4	SLOW
DI<4>	G15	IN	LVCMOS33	14	4	SLOW
DI<5>	C12	IN	LVCMOS33	14	4	SLOW
DI<6>	U14	IN	LVCMOS33	14	4	SLOW

Signal Name	Pin Number	Direction	IO Standard	IO Bank	Drive [mA]	Slew Rate
DI<7>	K15	IN	LVCMOS33	14	4	SLOW
DI<8>	T13	IN	LVCMOS33	14	4	SLOW
DI<9>	T15	IN	LVCMOS33	14	4	SLOW
DI<10>	T17	IN	LVCMOS33	14	4	SLOW
DI<11>	K17	IN	LVCMOS33	14	4	SLOW
DI<12>	T14	IN	LVCMOS33	14	4	SLOW
DI<13>	V12	IN	LVCMOS33	14	4	SLOW
DI<14>	V11	IN	LVCMOS33	14	4	SLOW
DI<15>	R15	IN	LVCMOS33	14	4	SLOW
DI<16>	V8	IN	LVCMOS33	14	4	SLOW
DI<17>	J5	IN	LVCMOS33	14	4	SLOW
DI<18>	K2	IN	LVCMOS33	14	4	SLOW
DI<19>	L5	IN	LVCMOS33	14	4	SLOW
DI<20>	D15	IN	LVCMOS33	14	4	SLOW
DI<21>	B10	IN	LVCMOS33	14	4	SLOW
DI<22>	D16	IN	LVCMOS33	14	4	SLOW
DI<23>	B15	IN	LVCMOS33	14	4	SLOW
DI<24>	E15	IN	LVCMOS33	14	4	SLOW
DI<25>	C16	IN	LVCMOS33	14	4	SLOW
DO<0>	F14	OUT	LVCMOS33	14	4	SLOW
DO<1>	C17	OUT	LVCMOS33	14	4	SLOW
DO<2>	D10	OUT	LVCMOS33	14	4	SLOW
DO<3>	D18	OUT	LVCMOS33	14	4	SLOW
DO<4>	H17	OUT	LVCMOS33	14	4	SLOW
DO<5>	G17	OUT	LVCMOS33	14	4	SLOW
DO<6>	V14	OUT	LVCMOS33	14	4	SLOW
DO<7>	L14	OUT	LVCMOS33	14	4	SLOW
DO<8>	R13	OUT	LVCMOS33	14	4	SLOW
DO<9>	N18	OUT	LVCMOS33	14	4	SLOW
DO<10>	V16	OUT	LVCMOS33	14	4	SLOW
DO<11>	U15	OUT	LVCMOS33	14	4	SLOW
DO<12>	P15	OUT	LVCMOS33	14	4	SLOW
DO<13>	T12	OUT	LVCMOS33	14	4	SLOW
DO<14>	U9	OUT	LVCMOS33	14	4	SLOW
DO<15>	U16	OUT	LVCMOS33	14	4	SLOW
DO<16>	M6	OUT	LVCMOS33	14	4	SLOW
DO<17>	R7	OUT	LVCMOS33	14	4	SLOW
DO<18>	P1	OUT	LVCMOS33	14	4	SLOW
DO<19>	J6	OUT	LVCMOS33	14	4	SLOW

Signal Name	Pin Number	Direction	IO Standard	IO Bank	Drive [mA]	Slew Rate
DO<20>	B16	OUT	LVCMOS33	14	4	SLOW
DO<21>	G14	OUT	LVCMOS33	14	4	SLOW
DO<22>	C11	OUT	LVCMOS33	14	4	SLOW
DO<23>	C14	OUT	LVCMOS33	14	4	SLOW
DO<24>	E13	OUT	LVCMOS33	14	4	SLOW
DO<25>	C13	OUT	LVCMOS33	14	4	SLOW
OE<0>	F17	OUT	LVCMOS33	14	4	SLOW
OE<1>	A17	OUT	LVCMOS33	14	4	SLOW
OE<2>	E17	OUT	LVCMOS33	14	4	SLOW
OE<3>	C18	OUT	LVCMOS33	14	4	SLOW
OE<4>	H16	OUT	LVCMOS33	14	4	SLOW
OE<5>	E16	OUT	LVCMOS33	14	4	SLOW
OE<6>	M17	OUT	LVCMOS33	14	4	SLOW
OE<7>	J14	OUT	LVCMOS33	14	4	SLOW
OE<8>	P18	OUT	LVCMOS33	14	4	SLOW
OE<9>	L18	OUT	LVCMOS33	14	4	SLOW
OE<10>	T18	OUT	LVCMOS33	14	4	SLOW
OE<11>	V17	OUT	LVCMOS33	14	4	SLOW
OE<12>	U11	OUT	LVCMOS33	14	4	SLOW
OE<13>	U12	OUT	LVCMOS33	14	4	SLOW
OE<14>	V9	OUT	LVCMOS33	14	4	SLOW
OE<15>	U10	OUT	LVCMOS33	14	4	SLOW
OE<16>	J4	OUT	LVCMOS33	14	4	SLOW
OE<17>	M5	OUT	LVCMOS33	14	4	SLOW
OE<18>	K3	OUT	LVCMOS33	14	4	SLOW
OE<19>	L3	OUT	LVCMOS33	14	4	SLOW
OE<20>	D14	OUT	LVCMOS33	14	4	SLOW
OE<21>	D13	OUT	LVCMOS33	14	4	SLOW
OE<22>	B14	OUT	LVCMOS33	14	4	SLOW
OE<23>	G16	OUT	LVCMOS33	14	4	SLOW
OE<24>	G15	OUT	LVCMOS33	14	4	SLOW
OE<25>	C12	OUT	LVCMOS33	14	4	SLOW

Table 4-3: Digital I/O Interface

4.3.1 TTL I/O Interface

The Speedgoat custom implementation will provide the requisite interfaces to drive the digital IO. The functionality described below is for user reference.

Each I/O line is buffered by a 74LVC2G241. The 74LVC2G241 is a tri-state buffer that provides TTL compatible inputs with 5 V-tolerance. The outputs can be set to tri-state with an output enable signal and provide a 47 Ω serial resistor and a 4.7 kΩ pull resistor. The pull resistor guarantees a valid logic level when the outputs are tristate and not driven externally. The pull voltage can be set to 3.3 V, 5 V or GND. When set to 5 V, the outputs must not be driven, but set to tristate to set a valid logical high.

A TVS array protects against ESD shocks.

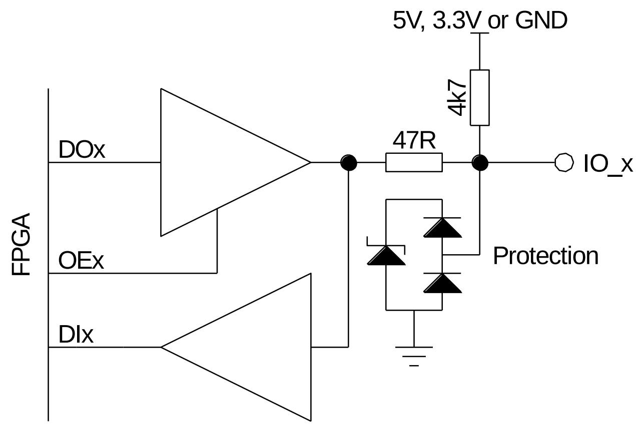


Figure 4-1: TTL I/O Interface

With the pull voltage set to 5 V, the digital I/O can weakly drive a higher voltage than 3.3 V by setting the output to tri-state. This can be useful when connecting to standard 5 V CMOS logic input, where a high level of minimum 3.5 V is required. Drive DO constant low and use OE to toggle the output.

With the pull voltage set to GND, a pull-down functionality is implemented. Drive DO constant high and use OE to toggle the output.

Pull Option	DO	OE	Output	Remark
No pull-up or pull-down	0	1	0	
	1	1	1	
Pull-up to 3.3 V	0	1	0	
	1	1	1	Driven to 3.3 V
	-	0	1	Pulled to 3.3 V
Pull-up to 5 V	0	1	0	
	-	0	1	Pulled to 5 V
Pull-down to GND	-	0	0	Pulled to GND
	1	1	1	

Table 4-4: I/O Pull Options

If the pull resistors float, the user should keep in mind that the I/O Lines are still connected via their pull resistors.

The normal behavior is that the User FPGA code controls the I/O Pull Configuration depending on User FPGA I/O Function. The SEL signals are connected to an analog multiplexer. With this multiplexer, the desired voltage can be adjusted directly from the User FPGA. The user must ensure that valid signals are always driven.

CNT Lines	Description	Artix-7 Pins
SEL[1:0]	11: pull-down 10: pull-up to 3.3 V 01: pull-up to 5 V 00: No pull-up or pull-down	V8, R2

Table 4-5: I/O Pull Configuration

4.3.1.1 Output Level & Output Current

Because of the 47 Ω series resistor, there is a reduced high-level voltage at the I/O pin when the output buffer sources a noticeable current to the external load while driving a high-level. To maintain a proper TTL high level, the recommended maximum I/O source current is 15 mA.

There is also an increased low-level voltage at the I/O pin when the output buffer sinks a noticeable current from the external load while driving a low-level. To maintain a proper TTL low level, the recommended maximum I/O sink current is 6 mA.

For achieving a 5 V CMOS high-level voltage (V_{OH} >= 3.5 V), the external load should be high impedance. If there would be a low impedance path to ground on the I/O load, this may result in a voltage divider with the on-board pull resistor, significantly reducing the high-level voltage at the I/O pin. To maintain a proper 5 V CMOS high, the I/O load (leakage) current should not exceed 250 µA.

4.4 Thermal Management

All components used on the IO391 are rated for the industrial temperature range of -40°C to +85°C. The actual temperature range the IO391 can be used in is highly dependent on the FPGA design, the load of the modules I/O circuitry and the applied cooling method.

The module has several hot spots including operational amplifiers, ADC and DAC, power supplies and the FPGA. The IO391 is equipped with a heatsink which will sink the module’s heat into the housing of the target machine, in the case of use in a Performance Realtime Target. The IO391 should not be powered without a heatsink mounted.

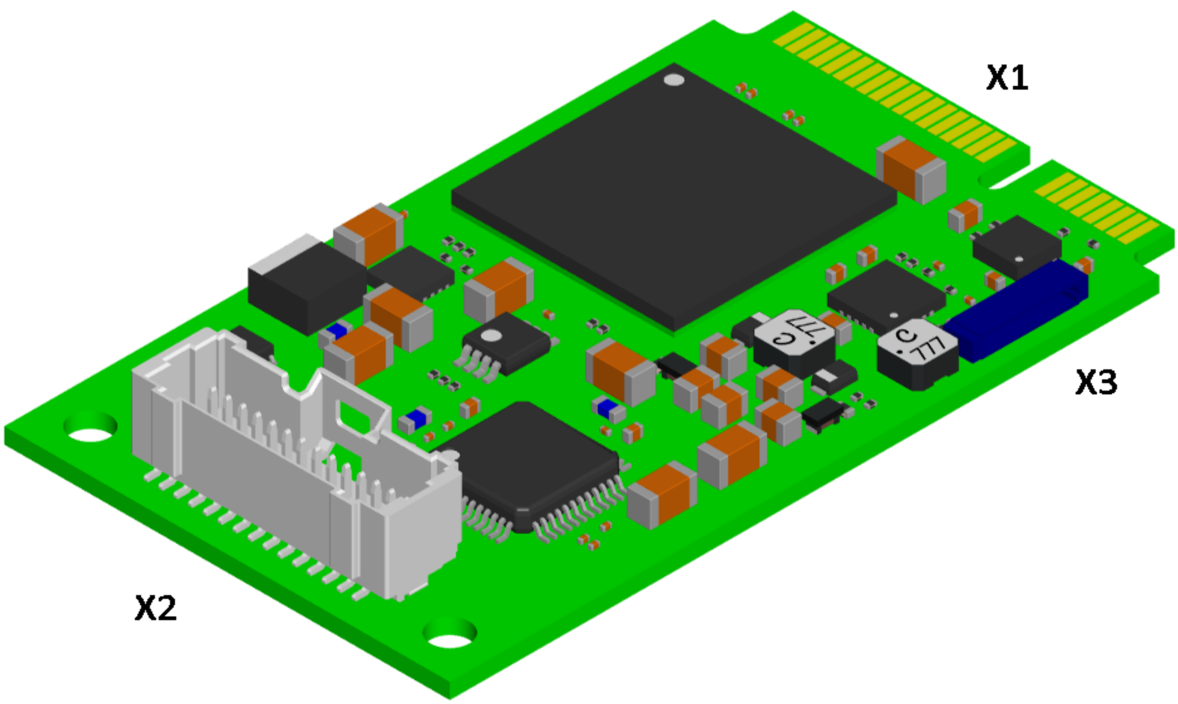
5 Installation

The IO391 module is supplied pre-mounted into your Realtime Target Machine.

6 I/O Interface

This chapter provides information about user accessible on-board connectors

6.1 Overview



X1	System Connector
X2	I/O Connector
X3	JTAG Connector

Figure 6-1: I/O Connector Overview

6.2 Board Connectors

6.2.1 System Connector (X1)

Pin-Count	52
Connector Type	Card-edge
Source & Order Info	none

Signal names in grey are not used by the card.

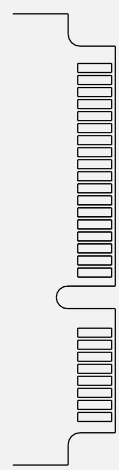
Description	Pin		Pin	Description
W_DISABLE2#	51		52	+3.3 Vaux
Reserved	49		50	GND
Reserved	47		48	+1.5 V
Reserved	45		46	LEP_WPAN#
GND	43		44	LEP_WLAN#
+3.3 Vaux	41		42	LEP_WWAN#
+3.3 Vaux	39		40	GND
GND	37		38	USB_D+
GND	35		36	USB_D-
PETp0	33		34	GND
PETn0	31		32	SMB_DATA
GND	29		30	SMB_CLK
GND	27		28	+1.5 V
PERp0	25		26	GND
PERn0	23		24	+3.3 Vaux
GND	21		22	PERST#
UIM_IC_DP	19		20	W_DISABLE1#
UIM_IC_DM	17		18	GND
Mechanical Key				Mechanical Key
GND	15		16	UIM_SPU
REF_CLK+	13		14	UIM_RESET
REF_CLK-	11		12	UIM_CLK
GND	9		10	UIM_DATA
CLKREQ#	7		8	UIM_PWR
COEX2	5		6	+1.5 V
COEX1	3		4	GND
WAKE#	1		2	+3.3 Vaux

Figure 6-2: Preliminary System Connector Pin Assignment

6.2.2 I/O Connector (X2) and M12 Front Panel Connector

Pin-Count	30
Connector Type	Molex Pico-Clasp, dual row straight header, with lock, with adapter to M12, front bezel
Source & Order Info	501190-3017
Mating Part	501189-2010

The I/O connector comes pre-connected to the front IO of your Baseline Realtime Target Machine. For the pinout of the M-12 front connectors for specific custom implementations, please refer to the documentation supplied with your custom implementation or HDLCoder integration package. The IO pin assignments below are for reference.

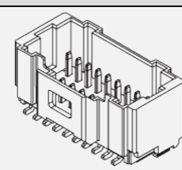
Description		**Pin Picoclasp**		**Pin**	**Description**	
Single-En.	**Diff.**				**Single-En.**	**Diff.**
GND	GND	1		2	GND	GND
I/O_0		3		4	I/O_1	
I/O_2		5		6	I/O_3	
I/O_4		7		8	I/O_5	
I/O_6		9	(Picoclasp)	10	I/O_7	
I/O_8		11		12	I/O_9	
I/O_10		13		14	I/O_11	
I/O_12		15		16	I/O_13	
I/O_14		17		18	I/O_15	
I/O_16		19		20	I/O_17	
I/O_18		21		22	I/O_19	
I/O_20		23		24	I/O_21	
I/O_22		25		26	I/O_23	
I/O_24		27		28	I/O_25	
GND	GND	29		30	GND	GND

Figure 6-3: I/O Internal Connector Pin Assignment

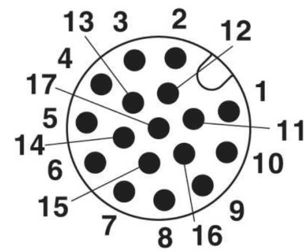
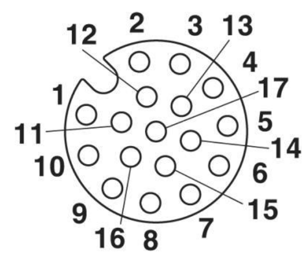
Connector B male		
Connector A female		

Figure 0-1: External M12 Connectors

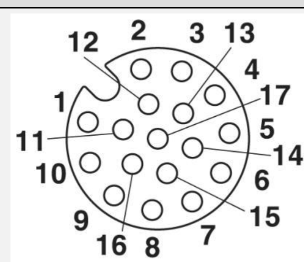
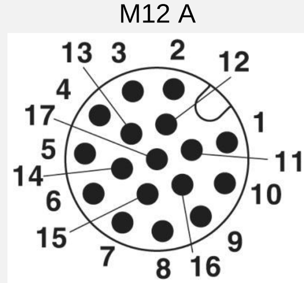
Description	Pin Phoenix M12		Pin Phoenix M12	Description
GND	A1		A2	GND
I/O_0	A3		A4	I/O_1
I/O_2	A5		A6	I/O_3
I/O_4	A7		A8	I/O_5
I/O_6	A9		A10	I/O_7
I/O_8	A11	M12 A	A12	I/O_9
I/O_10	A13		A14	I/O_11
I/O_12	A15		B3	I/O_13
I/O_14	B4		B5	I/O_15
I/O_16	B6		B7	I/O_17
I/O_18	B8		B9	I/O_19
I/O_20	B10		B11	I/O_21
I/O_22	B12	M12 B	B13	I/O_23
I/O_24	B14		B15	I/O_25
GND				GND
+5V				0V

Figure 0-2: M12 I/O Connector Pin Assignment

Note: 0V is internally connected to the ground of the target machine whereas analog ground is connected to the ground of the I/O module. 5V are supplied by the real-time target machine, which is capable of providing 1A continuous current. For more information, refer to the real-time target machine User Manual.

6.2.3 JTAG Connector (X3)

Pin-Count	10
Connector Type	JST XRS 10pol 0,6 mm Pitch IDC Connector
Source & Order Info	SM10B-XSRS-ETB
Mating Part	10XSR-36S

The IO391 provides a JTAG connector to access the FPGA's JTAG port.

Speedgoat provides a "Programming Kit" (TA308) which includes a XSR cable and an adapter module that provides a Xilinx USB Programmer II compatible 2 mm shrouded header. This kit can be used for debugging using the Xilinx toolchain.

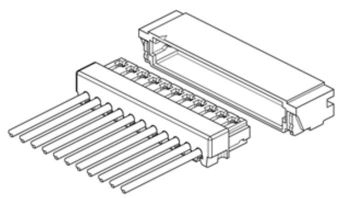
	Pin	Description
	1	GND
	2	TCK
	3	TMS
	4	TDI
	5	TDO
	6	GND
	7	GPIO0
	8	GPIO1
	9	PRESENT#
	10	V_{REF}

Figure 0-3: XRS Connector Pin Assignment

GPIO0 is connected to FPGA DONE

GPIO1 is connected to Power Good (covers FPGA V_{CORE} and 5 V)

PRESENT# is not used by the IO391.

V_{REF} is 3.3 V

Figure 0-4: IO391 connected to a Programmer via TA308